

1 A Tale of Two Compositions

This is going to be a long story. So, stick with me. Take it slow and complete each part in order. Took me a while to write this so it will take you some time to parse it!

1.1 Addresses

An organization typically keeps records of its employees' current and permanent addresses. Here's one way we can do this using classes:

```
class Employee {
    int    current_address_house_no;
    int    current_address_street;
    string current_address_city;
    string current_address_country;

    int    permanent_address_house_no;
    int    permanent_address_street;
    string permanent_address_city;
    string permanent_address_country;
};
```

However, as you can see, this is a lot of repetition. So, we realize that we can define a new class that is called `Address`. This class will have a few public information pieces – `house_no` and `street` (which will be integers) and `city` and `country` (which will be strings).

1. Your first task is to define this class `Address` with all these pieces of information (and they should be publicly visible).
2. Also, you need to define a publicly visible function (`get_full_address`) within this class that returns the full address in the following format. (Values are given as an example – `house_no` is 4, `street` is 25, `city` is Peshawar and `country` is Pakistan.)

H. No. 4, Street 25, Peshawar Pakistan

This function should *return a string* and *not* output the address line on the console! (You may wish to use the `to_string()` function to convert an `int` to `string`.)

After this, you should be able to run Part - 1 of the `main` function in the starter file.

1.2 Employees

Once we have the `Address` class ready, you can define an `Employee` class.

1. This class should have the employee's name and two other variables of type `Address`. One of these should be named `current_address` and the other `permanent_address`.

2. Also define a function (`set_current_address`) with the following declaration:

```
void set_current_address(int house_no, int street, string city, string country);
```

In the body of the function, you need to set the `house_no`, `street`, `city` and `country` of `current_address` variable to the values provided in the arguments. (Hint: you can use the dot operator to *access* or *assign* to the fields within the `current_address`).

3. Similar to the above, define another function as follows:

```
void set_permanent_address(int house_no, int street, string city, string country);
```

This would be exactly the same as above but will set the values within `permanent_address`.

4. Finally, write another function (`print_addresses()`) that prints both addresses **by calling the function `get_full_address()` you defined above for the `Address` class**. Do not try to re-write the code you've already written.

After this, you should be able to run Part - 2 of the main function in the starter file.

2 Wheels and Cars

This part is very similar to the first one. So, if you got that, this should be straight forward.

A Car is composed of four Wheel thingies. If a car is moving, all its wheels are moving (if nothing else has gone wrong).

2.1 Wheels for Cars

Let's first do wheels!

1. You need to define a `Wheel` class. At the moment, this class should have just a single variable state of type `string`. Make sure this variable is *hidden* from those outside the class!
2. Now, within this class, define a public function with the following declaration:

```
void set_wheel_state(string s);
```

This function should take set the state of the wheel to the value passed in as `s`.

3. Write another function in this class with the following declaration:

```
string get_wheel_state()
```

This should simply return the state of the wheel.

After this, you should be able to run Part - 3 of the main function in the starter file.

2.2 Cars on Wheels

Once we have a `Wheel` class, we can use it within our `Car` class.

1. Define a new class `Car`. This should have an *array of wheels* – the size of the array should be 4 since we have only four wheels. Make sure this array is *not* visible from outside the class.
2. Next, write a function `set_car_to_moving` within this class. This function should set the states of all four wheels to "Turning". That's it!
3. Write another function `set_car_to_stopped` same as above except the state of all four wheels should be set to "Stopped".
4. Finally, write a function `print_car_wheels_state` in this class which outputs the state of all four wheels in the following format:

Car state:

- Wheel 0 is Stopped
- Wheel 1 is Stopped
- Wheel 2 is Stopped
- Wheel 3 is Stopped

Of course, if the car is stopped, all output lines should have the word `Turning` there. Make sure you use the `get_wheel_state` function of the wheel you defined earlier for this.

After this, you should be able to run `Part - 4` of the `main` function in the starter file.

And you're done!

3 Submission

After you're done making your code work, write down (**read: assignment must be hand-written**) the full code you've created and submit that on or before the deadline. Late assignments (or soft copies) will not be accepted.